Dictionaries and records

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

Before you start

What does remaining(8, 3) return when its body returns capacity - used?

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Read and update named dictionary fields. Try the recall before opening notes.

Your learning target

I can read and update named **dictionary** fields.

I can connect a **record** to a function call.

I can distinguish missing fields from zero values.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Return values and local state. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Lists and indexing. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Boolean conditions. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Read and update named **dictionary** fields

Connect a **record** to a function call

Distinguish missing fields from zero values

Retrieve what you already know

> What does remaining(8, 3) return when its body returns capacity - used?

My starting explanation

This week's end goal

Trace dictionaries and **record**s and explain the changed program before running it.

Notice the evidence

> Investigate the supplied Dictionaries and records case. Separate observations, given facts and assumptions before choosing an explanation.

A fictional cupboard **record** has a name, capacity and used count. A list asks us to remember positions. Named fields make the intended meaning visible.

Our schema requires name as text and capacity and used as nonnegative integers, with used no greater than capacity.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Dictionaries and records. Connect each labelled input, deciding step and result.

```
cupboard = {"name": "Blue", "capacity": 12, "used": 4}
def remaining(capacity, used):
    return capacity - used

print(remaining(cupboard["capacity"], cupboard["used"]))
cupboard["used"] = 6
print(remaining(cupboard["capacity"], cupboard["used"]))
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Read used	4	12 - 4 = 8
Update used	6	12 - 6 = 6

Read each row from left to right and connect the evidence to the stated rule.

Square brackets look up the value for a key. The quoted key "used" is text; its value is an integer. Assigning to an existing key replaces that field. Other fields remain the same.

Dictionary syntax does not enforce our schema: the programmer must check it.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Dictionaries and records. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Change the Red **record** used to 9. State remaining spaces and whether capacity changes.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Add the field price with value 0.

Predicted result

Prediction: The price lookup now returns 0 instead of None.

Why the result follows

Reason: A **record**ed zero is a present value, whereas the earlier lookup had no field.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Dictionaries and records to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Dictionary	A mapping from unique keys to values

Word	Meaning
Key	A name used to look up a value
Record	Related fields describing one item
KeyError	The error from looking up an absent dictionary key

> **Use two terms to explain a step in this week's example.**

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Dictionaries and records, predict a step, then compare with its explanation.

> **Set used to 12. Predict the returned spaces. Does a used value of 0 mean the field is missing?**

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Dictionaries and records

A zero field is still present. A **dictionary record** describes a shelf. The program distinguishes missing data from an actual zero.

The facts we will use

Supplied facts

Python 3: shelf={"count":0,"name":"Atlas"}; print(shelf["count"]); print(shelf.get("price")); print("count" in shelf).

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Dictionaries and records

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read a stored field

count exists with numeric value 0.



2. Read an absent field

get("price") returns None because price is missing and no default was supplied.



3. Test key membership

"count" in shelf is True even though its value is zero.

CHANGE ONE THING

Add the field price with value 0.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read a stored field	count exists with numeric value 0.
2. Read an absent field	get("price") returns None because price is missing and no default was supplied.
3. Test key membership	"count" in shelf is True even though its value is zero.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Dictionary keys name fields. Presence is different from truthiness of a value; a measured zero should not be discarded as if the field were absent.

What this example does not establish

Keep the stated limit

None is also a possible explicitly stored value; real **record** contracts must define how missing and null values are represented.

Keep the stated limit

Read the labelled picture: Predict all three original **output**s. Explain the difference between count=0 and missing price, then predict the lookup after price=0 is added.

Keep the stated limit

Change one thing: Add the field price with value 0. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Dictionaries and records claim and explain a repair using the supplied evidence.

cupboard["Used"] raises **KeyError** here because keys are case sensitive. Correct it to "used". Adding a new field called "Used" would create a second key and leave the original unchanged; it would not repair the intended lookup.

Locate the earliest unsupported step

The code reads **record**["Used"] from the Red **record**. Explain the error and repair it without adding a field.

State one repair and the checks you would rerun.

Practise together

> Practise Dictionaries and records with support, then explain the deciding step in your own words.

For {"name": "Red", "capacity": 9, "used": 2}, state the keys, value of used and remaining spaces.

> **Set used to 12. Predict the returned spaces. Does a used value of 0 mean the field is missing?**

> **Explain your prediction to a partner before comparing results.**

Practise with less help: Trace an update

Use workbook W2 for the connected example “Dictionaries and **record**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Dictionaries and **record**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Predict all three original **output**s. Explain the difference between count=0 and missing price, then predict the lookup after price=0 is added.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Dictionaries and records. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Change the Red **record** used to 9. State remaining spaces and whether capacity changes.

```
cupboard = {"name": "Blue", "capacity": 12, "used": 4}
def remaining(capacity, used):
    return capacity - used
```

```
print(remaining(cupboard["capacity"], cupboard["used"]))
cupboard["used"] = 6
print(remaining(cupboard["capacity"], cupboard["used"]))
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Dictionaries and records. Explain why each step is justified.

Explain → apply → challenge

For {"name": "Red", "capacity": 9, "used": 2}, state the keys, value of used and remaining spaces.

The code reads **record**["Used"] from the Red **record**. Explain the error and repair it without adding a field.

A **record** has capacity 3 and used 4. It gives -1 spaces without an error. Should it be accepted? Explain.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Python readiness portfolio. Keep an annotated decision and the evidence supporting it.

Add a labelled dictionaries and **record**s **evidence** sheet to Python readiness portfolio. Show how you can read and update named **dictionary** fields.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this **record** to last week's **record**; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Named fields reduce position mistakes but do not prove that a **record** is accurate. A capacity of 2 with used of 5 is representable in a **dictionary** and violates this schema.

> **A record has capacity 3 and used 4. It gives -1 spaces without an error. Should it be accepted? Explain.**

A **record** has capacity 3 and used 4. It gives -1 spaces without an error. Should it be accepted? Explain.

> **My limited claim and a useful next check**

Study a complete case

> Revisit the complete Dictionaries and records case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Read used	4	12 - 4 = 8
Update used	6	12 - 6 = 6

Read each row from left to right and connect the evidence to the stated rule.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Read, update and pass a dictionary record

> Predict the three **output**s and trace the named-field update. Explain why zero and a missing count produce different results. If the condition becomes if not count, which supplied **record** is first handled incorrectly?

```
def describe(record):
    count = record.get("count")
    if count is None:
        return "missing"
    return count
record = {"count": 0}
print(describe(record))
record["count"] = 3
print(describe(record))
print(describe({}))
```

> **My independent explanation, trace and limits**

Attempt this case before feedback. **Record** any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Create a Green **record** with capacity 15 and used 6. Pass its fields to remaining and store the result. Update used to 10 and repeat. Predict both results.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. **Record** whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Dictionaries and records, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A zero field is still present. A **dictionary record** describes a shelf. The program distinguishes missing data from an actual zero.

By the end, I can	How I will show it
Read and update named dictionary fields	Explain the deciding step, show a trace or test, and state one limit.
Connect a record to a function call	Explain the deciding step, show a trace or test, and state one limit.
Distinguish missing fields from zero values	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Dictionaries and records without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Read and update named **dictionary** fields

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A zero field is still present. A **dictionary record** describes a shelf. The program distinguishes missing data from an actual zero.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? "count" in shelf is True even though its value is zero.

1. Test key membership

2. Read an absent field

3. Read a stored field

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or **record**s it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Test key membership "count" in shelf is True even though its value is zero. **Dictionary** keys name fields.

Presence is different from truthiness of a value; a measured zero should not be discarded as if the field were absent.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Add the field price with value 0. Which conclusion is supported by the supplied facts?

1. The price lookup now returns 0 instead of None.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The price lookup now returns 0 instead of None. A **record**ed zero is a present value, whereas the earlier lookup had no field.

Repair a missing building block

Review Return values and local state (python-bridge/9). Goal: Use a returned value in a later calculation. Short practice: Trace the two returned values and explain why first stays seven.

What happens if the caller tries to print result directly?. Return to this week and explain what you changed.

Review Lists and indexing (python-bridge/7). Goal: Distinguish a list index from the value stored at that position. Short practice: Label the three index positions.

Predict all original **output**s, then name the first **output** and stopping point after using index 3.. Return to this week and explain what you changed.

Review Boolean conditions (python-bridge/4). Goal: Translate stated requirements into comparisons. Short practice: Give both comparison results for ages 11, 15 and 16.

Which comparison rejects each outside value, and why does and matter?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

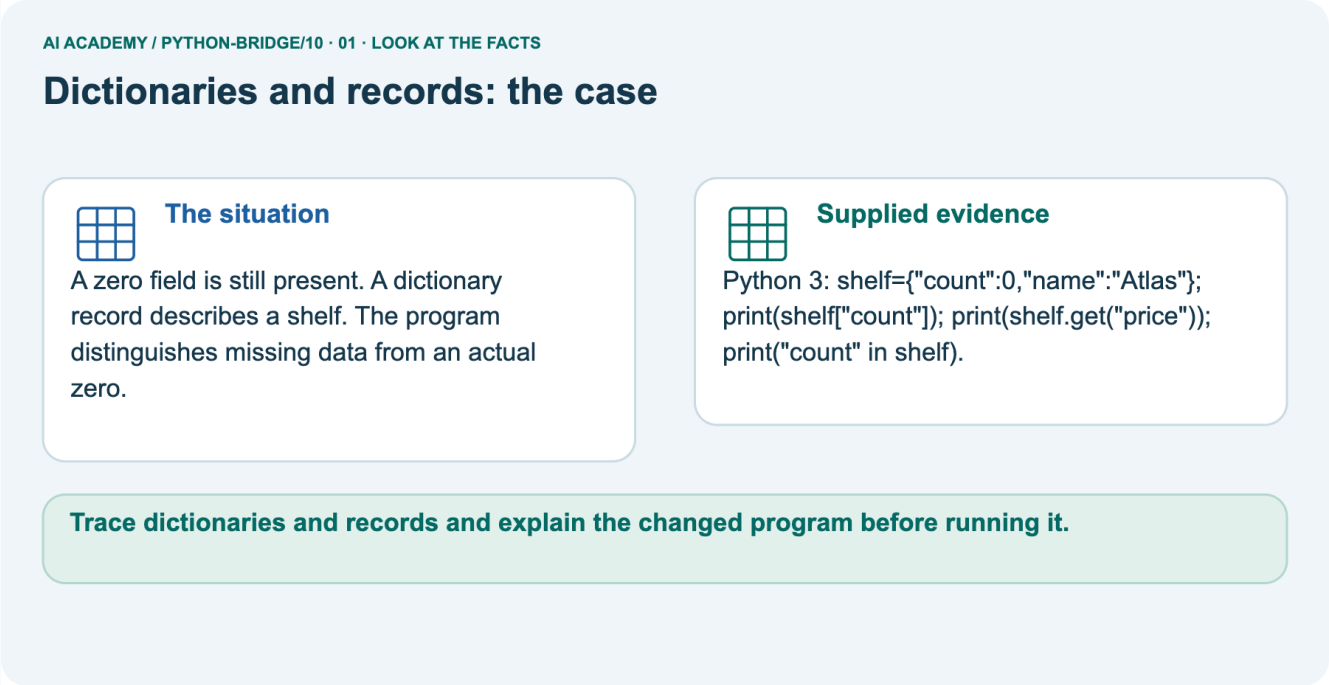
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Dictionaries and records graphic using its labels, deciding evidence and relevant limitation.

01. Dictionaries and records: the case



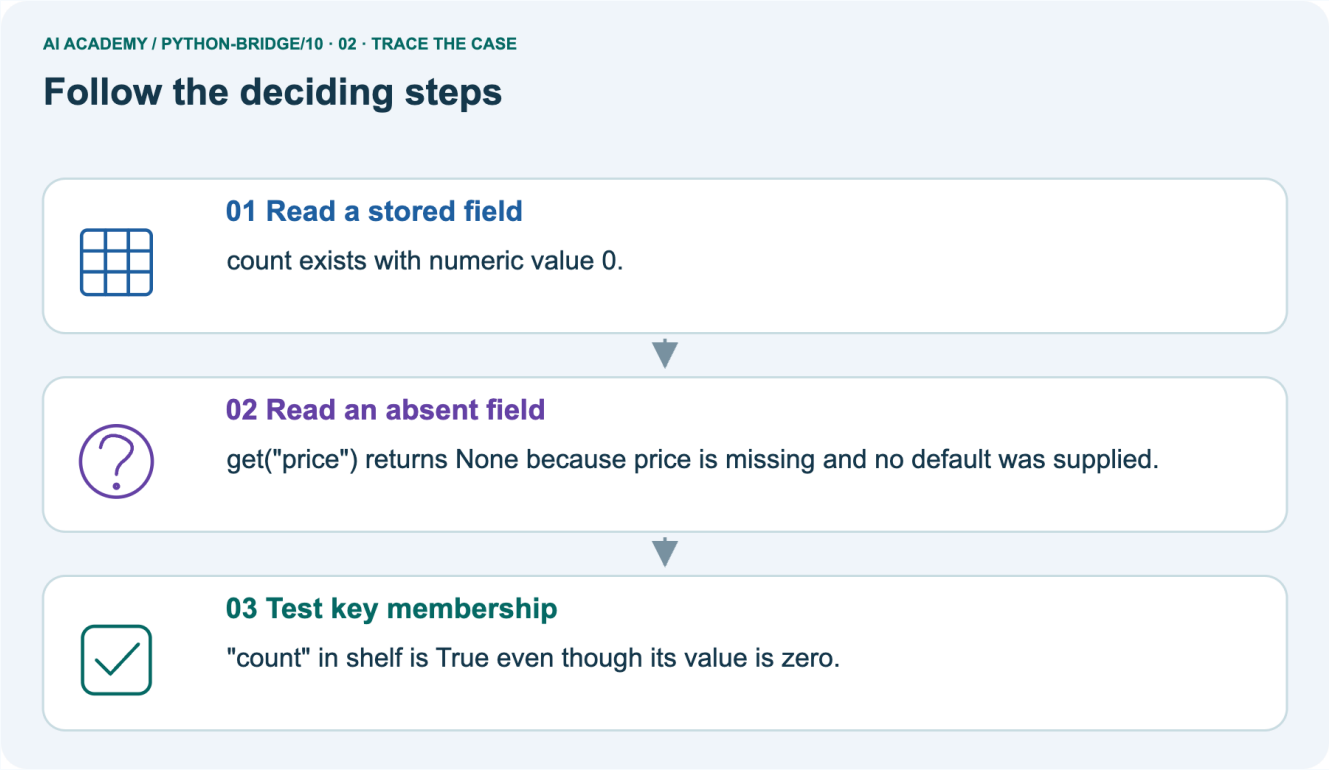
Follow the labelled picture with the matching explanation. Open the vector diagram

A zero field is still present. A **dictionary record** describes a shelf. The program distinguishes missing data from an actual zero.

Supplied facts: Python 3: shelf={"count":0,"name":"Atlas"}; print(shelf["count"]); print(shelf.get("price")); print("count" in shelf).

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read a stored field: count exists with numeric value 0. → Read an absent field: get("price") returns None because price is missing and no default was supplied. → Test key membership: "count" in shelf is True even though its value is zero.

> Point to each numbered stage and explain which **record**ed **input** it uses.

03. Follow the evidence

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Follow the evidence

Step or case	Evidence	Meaning
Read used	4	12 - 4 = 8
Update used	6	12 - 6 = 6

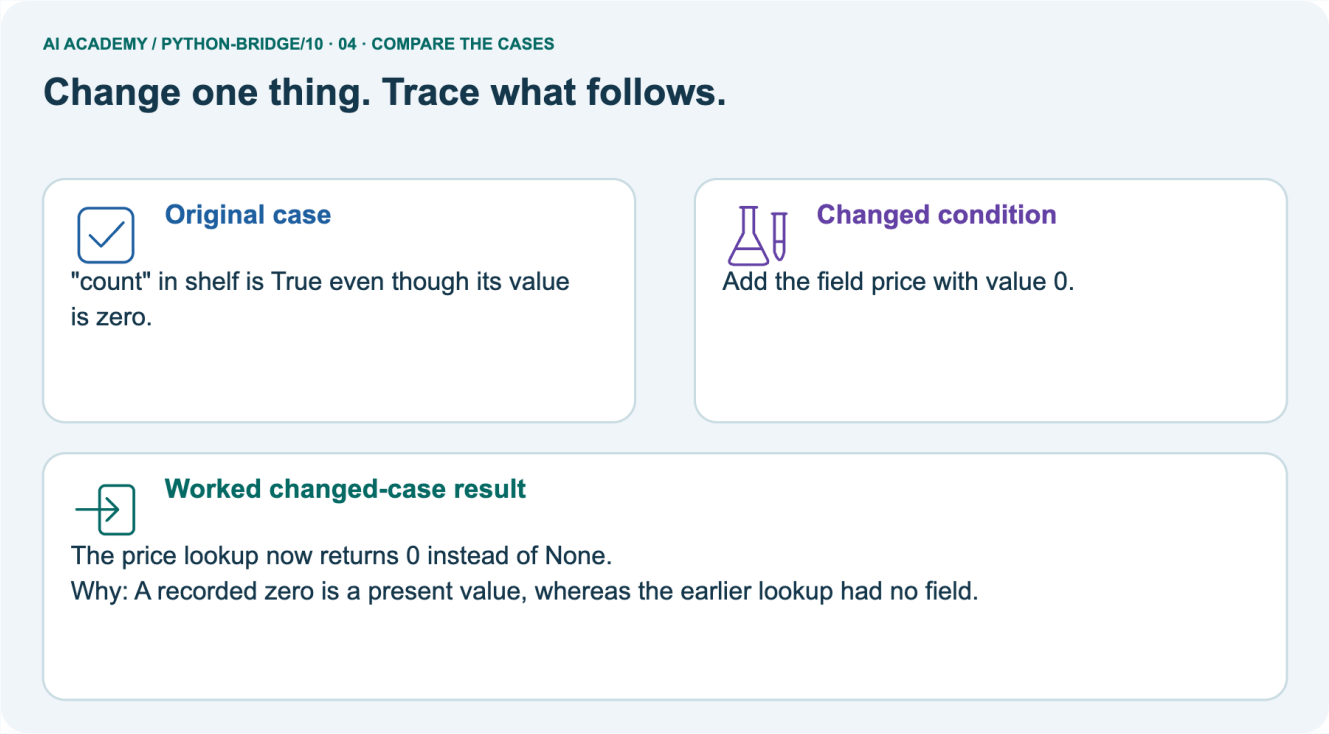
Read each row from left to right and connect the evidence to the stated rule.

Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Read used | 4 | 12 - 4 = 8. Update used | 6 | 12 - 6 = 6

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



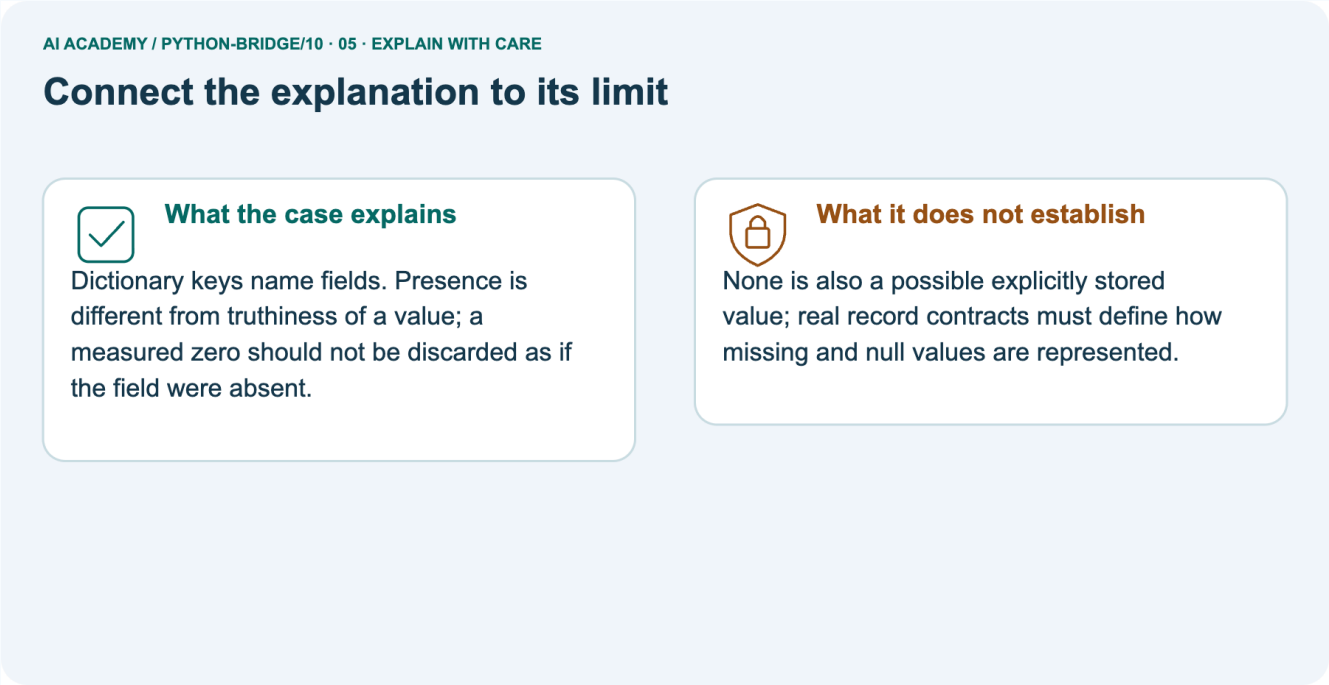
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: "count" in shelf is True even though its value is zero. Changed condition: Add the field price with value 0. Expected result: The price lookup now returns 0 instead of None.

Reason: A **record**ed zero is a present value, whereas the earlier lookup had no field.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



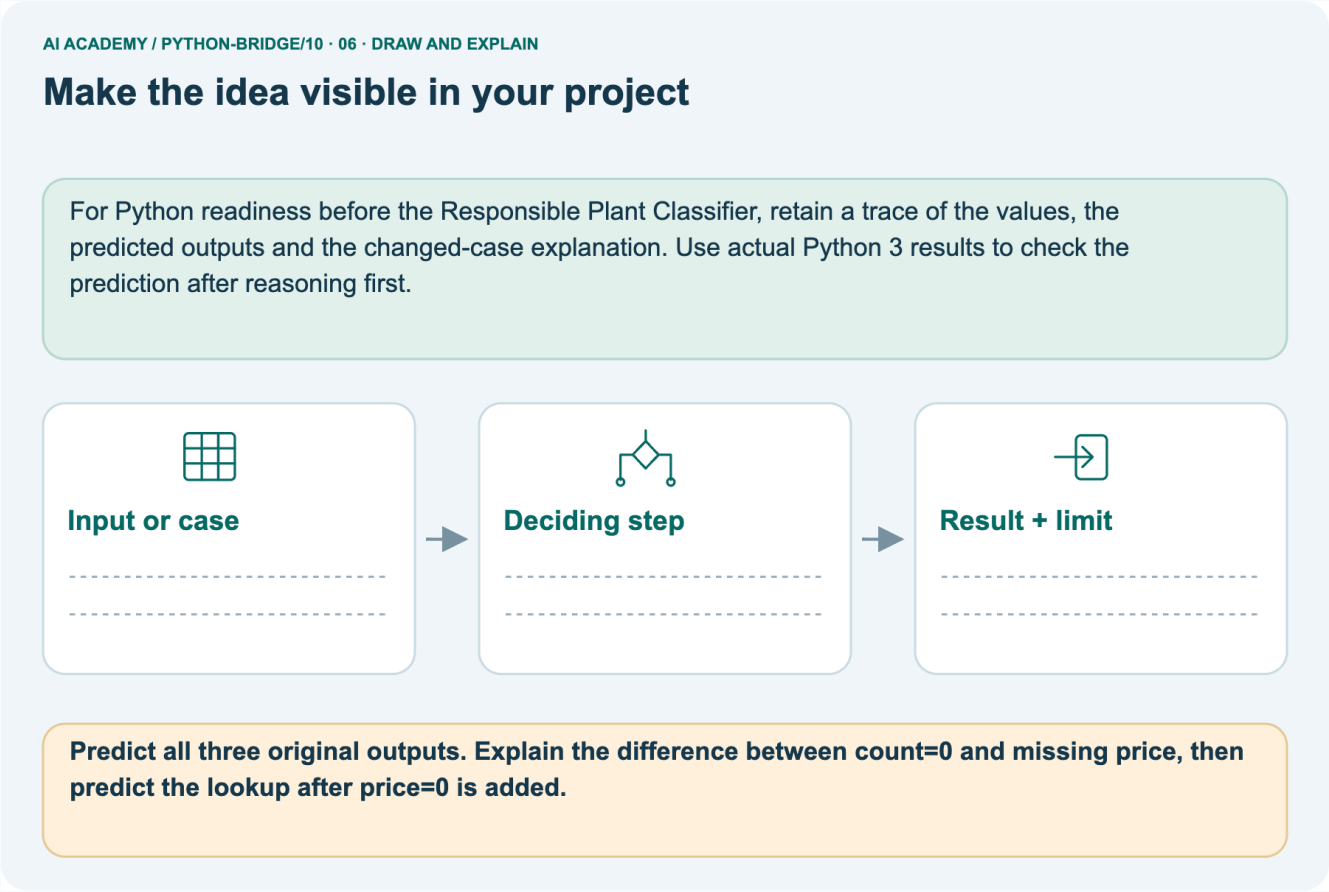
Follow the labelled picture with the matching explanation. Open the vector diagram

Dictionary keys name fields. Presence is different from truthiness of a value; a measured zero should not be discarded as if the field were absent.

Limit: None is also a possible explicitly stored value; real **record** contracts must define how missing and null values are represented.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Predict all three original **output**s. Explain the difference between count=0 and missing price, then predict the lookup after price=0 is added.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.